

Tasks 1: Assessing utility function (application to plastics sorting problem)

Identify the **utility matrix** corresponding to **plastics sorting problem** where 4 classes are considered {ABS, HIPS, PE, PP}. **(5 points)**

Tasks 2: Supervised classification using expected utility (application to plastics sorting problem)

In this second task, the aim is to **build a new classifier** that implements the **expected utility function**. First a **classical classification algorithm** is considered to obtain **posterior probabilities** then the **expected utility model** is used to rank the classes. For a given new example, the prediction corresponds to the class obtaining the maximum expected utility.

To do so one has to perform the following steps applied to the plastics sorting problem:

- a. Implement the **expected utility model**. **(2 points)**
- b. Test your algorithm using **several classical prediction algorithms** on the validation data of plastics (split your data into train and validation). **(2 points)**
- c. highlight some examples where the behaviour of the new algorithm corresponds to the expected results of the DM. **(1 points)**

Tasks 3: Supervised classification using evidential KNN (application to plastics sorting problem)

The same problem as in the task 3 but using **eknn classifier**:

- a. Implement eknn classifier. **(6 points)**
- b. Implement the **expected utility model** with eknn classifier. **(2 points)**
- c. Test your algorithm on the **validation data** of plastics (split your data into train and validation). **(1 points)**
- d. highlight some examples where the **classic knn algorithm** gives different results to those of eknn. **(1 points)**